INCH-POUND

MIL-C-24643/36C
14 March 1994
SUPERSEDING
MIL-C-24643/36B
3 October 1986

MILITARY SPECIFICATION SHEET

CABLE, ELECTRICAL, TYPE LS3SWU
(INCLUDING VARIATION LS3SWA AND LS3SWUS)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-C-24643.

This cable can be manufactured in three variations: LS3SWU (unarmored), LS3SWA (armored) and LS3SWUS (double overall shield).

REQUIREMENTS:

Qualification required.

Construction (watertight)

First - Copper conductor, AWG no 18 (7 by 0.0152 inch), uncoated. Maximum diameter 0.050 inch.
Second - Separator may be used at manufacturer's option where required to provide free-stripping insulation.
* Third - Crosslinked polyethylene insulation. Nominal wall thickness 0.023 inch; wall thickness may vary from nominal as necessary, in order for completed cable to meet the specified electrical requirements. Colored insulation, one black, one white, and one red conductor for each triad.
Fourth - Three conductors (one black, one white and one red) cabled together with a nominal lay of 3.5 inches to form each triad.
Fifth - Binder tape over each triad, applied at manufacturer's option, applied helically with overlap.
Sixth - Braided shield of tin coated copper. Minimum coverage of 85 percent.

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Construction (non-watertight) (continued)

*	Seventh	- Shield insulation of two polyester tapes, type G in accordance with MIL-I-631, outer tape sealed, An alternate shield insulation may consist of one tape, type G of MIL-I-631, plus an extruded jacket of a suitable transparent material, minimum average thickness of the transparent material 0.003 inch. The standard identification code shall be applied by method 2 on the inner tape. Diameter over completed triad 0.240 inch nominal.
*	Eighth	- The required number of triads (see table I) cabled together with a lay not greater than 24 times the pitch diameter of the layer. Cabling sequence shall be consecutive, starting with no. 1, from the center outward. Fillers shall used as necessary to form a firm, well rounded assembly.
	Ninth	- Binder tape, applied helically with overlap.

Construction variant for type LS3SWU

*	Tenth	- Cross-linked polyolefin jacket. (see table I for thickness).

Construction variant for type LS3SWU

*	Tenth	- Cross-linked polyolefin jacket. (see table I for thickness).
*	Eleventh	- Braided metal armor

Construction variant for type LS3WSUS

*	Tenth	- Braided shield of tin coated copper. Minimum coverage of 85 percent.
	Eleventh	- Spirally wrapped 0.002 inch thick polyethylene terephthalate (mylar) Separator tape with 25 percent minimum overlap.
*	Twelfth	- Braided shield of tin coated copper. Minimum coverage of 85 percent.
	Thirteenth	- Binder tape applied helically with overlap.
*	Fourteenth	- Cross-linked polyolefin jacket. (see table I for thickness).

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TABLE I. Details.

Military 1/ part no. M24643/36	Size 2/	Number of triads	Cable jacket thickness min avg (inch)	Diameter over Shielded core (inch nom.)	Overall diameter (inches)					
					TYPE LS3SWU		TYPE LS3SWA		TYPE LS3SWUS	
					min	max	min	max	min	max
-01UN	3	3	0.040	0.561	0.610	0.655	0.665	0.705	0.675	0.710
-02UN	7	7	.040	.759	.880	.940	.930	.990	.940	1.000
-03UN	10	10	.050	1.024	1.100	1.180	1.150	1.230	1.160	1.240
-04UN	14	14	.050	1.080	1.200	1.280	1.250	1.330	1.260	1.340
-05UN	19	19	.050	1.243	1.370	1.450	1.420	1.500	1.430	1.510
-06UN	24	24	.050	1.345	1.640	1.760	1.690	1.810	1.700	1.820
-07UN	30	30	.050	1.430	1.760	1.860	1.810	1.910	1.820	1.920
-08UN	37	37	.060	1.560	1.890	1.940	1.910	2.010	1.950	2.050
-09UN	44	44	.060	1.780	2.140	2.240	2.190	2.290	2.200	2.300

1/ Unarmored. When armor option or double overall shield is required, see MIL-C-24643 for configuration of part number.
2/ Type LS3SWU, LS3SWA and LS3SWUS.

EXAMINATION AND TESTS:

	Requirements
Basic electrical:	
* Conductor Resistance - ohms/1000 feet at 25°C, maximum	7.06
Voltage withstand - volts, root mean square, minimum	
Conductor to conductor	2000
Shield to shield (Triad shields only)	500
* Shield to shield (LS3SWUS overall shield only)	200
Insulation resistance - Megohms/1000 feet, minimum	
* Conductor to shield	500
* Component shield to shield	100
* Shield to shield (LS3SWUS overall shield only)	100
Jacket flaws	No failure
* Conductor and shield continuity	No failure
Group A:	
Visual and dimensional	No failure
Hydrostatic (open end) - leakage at 300 lb/in^2, 2 hours, in^3, maximum	1.0
Capacitance	
Mutual capacitance - 1 kilohertz (KHz), pF/ft, maximum	30
* Capacitance unbalance - percent maximum	10

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EXAMINATION AND TESTS:	Requirements
Group B:	
Cross-linked proof test (percent, maximum)	
Insulation	50
* Jacket (When tested at 200°C)	50
Tear - pounds per inch thickness, minimum (ASTM D 470)	35
Physicals (unaged)	
Insulation	
Tensile strength - lb/in^2, minimum	1800
Elongation - percent, minimum	250
Jacket (cable)	
Tensile strength - lb/in^2, minimum	1300
Elongation - percent, minimum	160
Group C:	
Physicals (aged)	
Insulation	
Air oven	
Tensile strength - percent of unaged, minimum	80
Elongation - percent of unaged minimum	80
Jacket (cable)	
Air oven	
Tensile strength - percent of unaged, minimum	60
Elongation - percent of unaged minimum	60
Hot oil immersion	
* Tensile strength - percent of unaged, minimum	50
* Elongation - percent of unaged, minimum	50
Shrinkage	No failure
Heat distortion - percent of unaged, maximum	30
* Permanence of printing (jacket) - (type LS3SWU and LS3SWUS only) cycles, minimum	125
* Cable sealant removability	No failure
Armor (type LS3SWA only) - conformance to material, construction and coverage	No failure
Shield - conformance to material, construction and coverage	No failure
* Surface transfer impedance (type LS3SWUS only)	
Milliohms per meter, maximum	100
Emp response time - db, minimum	60
* Group D:	
Flame propagation (cable)	No failure

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QUALIFICATION INSPECTION:

* Qualification inspection shall include basic electrical, all of groups A, B, C and D, plus the following:

	Requirements
Aging and compatibility (cable) (125± 5°C.)	No failure
Abrasion resistance (jacket) - scrapes, minimum	75
Acid gas equivalent - percent, maximum	
Jacket	2
Fillers	2
Insulation	18
* Shield insulation	2
Halogen content - percent, maximum	
Jacket	0.2
Fillers	0.2
Immersion (jacket)	
Tensile strength - percent of unaged, minimum	50
Elongation - percent of unaged, minimum	50
Smoke index, maximum	
Jacket	25
* Fillers	45
Insulation	45
Toxicity index, maximum	
Jacket	5
Fillers	5
Insulation	1.5
Durometer (jacket) - (type A) hardness, minimum	80
Weathering (jacket)	No failure

UNIT ORDERING LENGTHS:

Size	Feet (nominal)
3 through 19	1000
24 through 44	500

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Navy - SH

Review Activities:
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Navy - EC, CG
DLA - IS

Preparing Activity:
Navy - SH

Agent:
DLA-IS

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